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# Reinforcement learning for QoS-guaranteed intelligent routing in Wireless Mesh Networks with heavy traffic load

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**Abstract**

Wireless Mesh Networks is increasingly being applied widely with explosive traffic demand. This leads to a great challenge for traditional routing protocols in ensuring Quality of Service. We propose a QoS-guaranteed intelligent routing algorithm in this paper for WMN with heavy traffic load using reinforcement learning to improve its performance. We build a reward function for the Q-Learning algorithm to choose a route so that the packet delivery ratio is the highest. Concurrently, the learning rate coefficient is flexibly changed to determine constraints of the end-to-end delay. Our performance evaluations show that the proposed algorithm has significantly improved performance compared with other well-known routing algorithms.
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*Keywords:* Reinforcement learning; Wireless Mesh Networks; QoS-guaranteed routing; Intelligent routing

## 1. Introduction

Wireless Mesh Networks (WMN) [1,2] is growing and increasingly being used in many fields, especially in the trend of the digital transformation of society nowadays. For the explosion of traffic demand due to the digital transformation in offices, enterprises, schools, and so on, most WMNs are under heavy traffic load. As a result, the probability of congestion can increase greatly, leading to an increase in end-to-end delay (EED), bit error rate, and reduce throughput. This greatly affects the QoS in the network. Therefore, it is necessary to study routing protocols to improve performance and ensure QoS for the WMN with heavy traffic load.

The improvement of routing protocols in WMNs can be done by many different methods. A common method used by research groups is to build a routing metric taking into account performance parameters, and then choose a route with the best metric. HopCount is the simplest routing metric, used in traditional routing protocols such as Dynamic Source Routing Protocol (DSR) [3], and Ad hoc On-Demand Distance Vector Routing (AODV) [4]. In the case of WMNs with high traffic load, the HopCount metric may not be suitable because it can cause bottlenecks in the network. In addition, the HopCount metric does not consider transmission interference and link quality. Therefore, it is not suitable for QoS-guaranteed routing techniques. One routing metric that takes into account link quality commonly used in WMNs is the Expected Transmission Count (ETX) [5]. The ETX of a link is calculated by the forward and reverse delivery ratios of that link. A routing metric that is improved from ETX is the Expected Transmission Time (ETT) [6], where the packet size and the link capacity are added to the ETX metric. Another routing metric that takes into account the signal to noise ratio (SNR) of links is proposed by the authors of [7], called Weighted Signal to noise ratio Average (WSA) for dynamic sequence distance vector (DSDV) routing protocol. The WSA metric uses the SNR parameter provided by the physical layer based on the cross-layer model.

The routing metrics mentioned above are often applied to traditional routing protocols [5–7]. Recently, machine learning methods have been applied to routing algorithms, with the most common using reinforcement learning (RL) [8–10]. For this method, the Q-Learning algorithm is often used to choose

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the route, where the routing metric is used as a reward in the learning process to build a routing table. This is an intelligent routing technique that is suitable for network models where the routing algorithm has to do complex and frequent calculations to determine the optimal route.

In this paper, we focus on investigating the application model of the RL for routing problems in WMNs. Then, an intelligent routing algorithm is proposed to improve the performance of the WMNs in case of heavy traffic load. The new contributions of this work are summarized as follows: (i) We propose a reward function for the Q-Learning algorithm that reflects the packet delivery probability (PDP) over a route from source to destination in the WMNs. (ii) We propose a method to determine the constraint of the EED in the WMNs by flexibly changing the learning rate coefficient of the Q-Learning algorithm. (iii) We propose a mechanism to self-learn the routing information at each node according to each traffic load update period. This allows minimizing the overhead in the network. (iv) We propose an intelligent routing algorithm namely QGIR (QoS-guaranteed Intelligent Routing) using RL for the WMNs with heavy traffic load.

The next sections of this paper are organized as follows. Section 2 analyzes the application model of reinforcement learning for routing in WMNs and published works related to this topic. Section 3 presents our proposed routing algorithm. Section 4 presents the simulation results and discussion. Finally, concluding remarks and promising future study items are given in Section 5.

## 2. Application of the RL for routing in WMNs

The RL is a type of machine learning concerned with how a system can learn from its previous actions with its environment to choose actions in the future efficiently. The essence of the RL is trial-and-error, which means repeating the trials and learning from each trial. The basic principle of the RL is that an agent acts as a learner, interacting with its environment to select an action so that the obtained rewards are the greatest. Considering an example of the routing problem in the WMN, in this context, a wireless router (WR) interacts with its neighbors to learn a route to the destination node. Each time WR interacts with a neighbor node, the obtained reward is the total cost to the destination. Through the learning process, WR will select the neighbor node to get the highest rewards, in the sense that the cost to go to the destination is the best.

In the RL method, the total reward after performing action $a_t$ at state $s_t$ is $Q(s_t, a_t)$, determined by Q-Learning is a RL algorithm. $Q(s_t, a_t)$ is called Q-value, updated according to (1):

$$Q(s_t, a_t) = (1-\alpha) \times Q(s_t, a_t) + \alpha[R(s_{t+1}, a_{t+1}) + \gamma \times \max_{\forall a_{t+1}} Q(s_{t+1}, a_{t+1})] \quad (1)$$

where $\alpha$ and $\gamma$ are the learning rate and the discount factor, respectively. $\alpha$ and $\gamma \in [0, 1]$. The Q-Learning has been successfully applied for routing in wireless networks [8–10]. The Q-value in (1) is often used as the objective of the routing algorithms. With the goal of choosing the best route to the gateway, the authors of [9] have proposed an optimal routing algorithm namely RLBPR (RL-based Best Path Routing). In this algorithm, the authors have defined a link quality metric (LQM) as a function of the IR and CL. Then, a formula for calculating the path quality (PQ) to the gateway is defined based on the LQM. The PQ is used as the reward of the RL algorithm for selecting the route to the gateway. In [11], the authors have proposed two route selection schemes based on RL to improve the performance of the multi-hop cognitive radio network. These schemes are named traditional RL and RL-based with average Q-value. Both schemes use the available channel time at the bottleneck link as the reward of the Q-Learning. This metric is used for selecting the route between a pair of source–destination nodes. Also using the RL method, the authors of [12] have proposed an energy balanced routing protocol based on Q-learning (QEBR) for WMN. The QEBR is executed according to the principle of the distributed routing. The authors proposed the concept of neighbor energy sorting to use for the reward of the Q-learning algorithm.

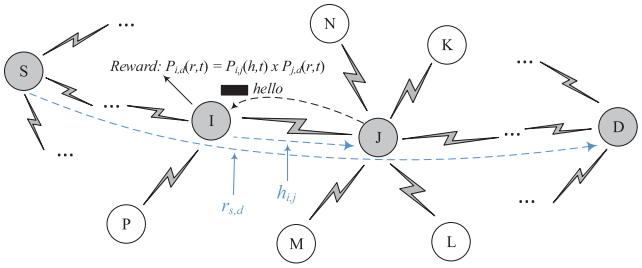



**Fig. 1.** The model of updating the Q-value for QGIR algorithm.

Through some of the published works mentioned above, we have commented that the RL is an efficient solution for routing in wireless networks. The proposed algorithms based on the RL achieve high efficiency. In this paper, we also use the ML to propose an intelligent routing algorithm for the WMNs in the case of heavy traffic load. Our idea is to use the Q-value to store the packet delivery probability (PDP) from source to destination. Concurrently, the coefficient of the learning rate is flexibly changed to determine the constraints of the EED in the WMNs.

## 3. Our proposed routing algorithm

In this section, we propose the QGIR algorithm for WMNs with heavy traffic load using RL. The QGIR algorithm aims to increase the PDP and the EED is always guaranteed. This is done by the reward function and learning rate coefficient of Q-Learning algorithm. We first list the important notation used for the representation of QGIR algorithm which is shown in Table 1.

### 3.1. PDP

Considering a multi-hop route from source node S to destination node D in the WMN as shown in Fig. 1 ($r_{s,d}$), according to statistical probability theory, the PDP of a data packet that

**Table 1**
Description of the notation used for QGIR algorithm.

| Notation | Description |
|---|---|
| $r_{s,d}$ | Route from the source ($S$) to the destination ($D$) |
| $h_{i,j}$ | Hop from the node $I$ to the node $J$ |
| $P_{s,k,d}(r,t)$ | PDP of a data packet that travels the route $r_{s,d}$ during the period $t$ with $k$ is the next node |
| $P_{i,j}(h,t)$ | PDP of a data packet that travels the hop $h_{i,j}$ during the period $t$ |
| $N_s(t)$ | Number of the data packets that the node $I$ sends to the node $J$ during the period $t$ |
| $N_r(t)$ | Number of the ACK packets that node $I$ is successfully received from node $J$ during period $t$ |
| $P_{i,j}(h,l)$ | PDP of the hop $h_{i,j}$ that node $I$ learned to update its routing table |
| $P_{j,k,d}(r,l)$ | PDP of the route $r_{j,d}$ that node $I$ learned to update its routing table |
| $R_{j,k,d} =$ | $\{D, K, P_{j,k}(h), P_{j,k,d}(r)\}$: The route element from node $J$ to node $D$ with node K as the next node and $P_{j,k,d}(r)$ as the metric |

travels the route $r_{s,d}$ during a period of time $t$, $P_{s,d}(r,t)$, is given by

$$P_{s,d}(r,t) = \prod_{\forall h_{i,j} \in r_{s,d}} P_{i,j}(h,t) \tag{2}$$

where $P_{i,j}(h,t)$ is the PDP of a data packet that travels the hop from node $I$ to node $J$ ($h_{i,j}$) during a period $t$. $P_{i,j}(h,t)$ is measured by

$$P_{i,j}(h,t) = N_s(t)/N_r(t) \tag{3}$$

where $N_s(t)$ and $N_r(t)$ are the number of data packets which node I sends to node J and the number of ACK packet which node I is successfully received from node J during a period $t$, respectively. Same as [8], to avoid fluctuations, we use a coefficient $\lambda \in [0, 1]$ that represents moving average for the $N_s(t)$ and $N_r(t)$. Thus, these parameters are determined as follows

$$N_s(t) = \lambda N_s(t) + (1-\lambda)N_s(t-1) \tag{4}$$

$$N_r(t) = \lambda N_r(t) + (1-\lambda)N_r(t-1) \tag{5}$$

### 3.2. Reward function

For the routing algorithms based on the RL, the reward function acts as the routing metric. Because the objective of the QGIR algorithm is to increase the PDP, we use the PDP metric of hops, determined according to (3) as the reward of the RL algorithm.

### 3.3. Updating Q-value in the QGIR algorithm

The RL model for QGIR algorithm is illustrated in Fig. 1. When node I receives a Hello packet from node J which is its neighbor, node I will learn to update its routing information from it to the destination with J being the next node. Based on the PDP equation of a route (e.q (2)), we modify the basic equation of the RL to obtain Q-value that is the PDP of the route from node I to the destination node, updated according to (6):

$$Q(i,j,d) = (1-\alpha) \times Q(i,j,d) + \alpha[R_{i,j}(h,t) \times \gamma \times \max_{\forall k \in L_j} Q(j,k,d)] \tag{6}$$

where $Q(i,j,d)$ is the PDP on the route from node I to the destination node (D) and J is the next node ($I \rightarrow J \rightarrow ... \rightarrow D$). $R_{i,j}(h,t)$ is the reward of the action that chooses J as the next node to go to the destination. It is also the PDP of the hop $h_{i,j}$ in our context, i.e. $R_{i,j}(h,t)$ is equal to $P_{i,j}(h,t)$, determined according to (3). $L_j$ is the set of the neighbors of the node J. The factors of $\alpha$ and $\gamma$ are the learning rate and the discount, respectively. At any node I, when choosing a route to transmit data, the route with the maximum Q-value that is determined according to (13) is chosen, that is, the PDP from node I to the destination node is the highest.

### 3.4. Determine the constraint of the EED

To ensure QoS in the WMN, the EED must be within the allowable limits. In our proposed algorithm, the EED is always constrained during learning to update the routing table. When a packet $p$ is transmitted through the route $r_{s,d}$, the EED is measured by

$$T^{(p)}_{s,d}(r) = \sum_{\forall h_{i,j} \in r_{s,d}} T^{(p)}_{i,j}(h) \tag{7}$$

where $T^{(p)}_{i,j}(h)$ is the EED of the packet $p$ when it transmits through the hop $h_{i,j}$. $T^{(p)}_{i,j}(h)$ is given by

$$T^{(p)}_{i,j}(h) = \tau^{(trans)}_{i,j} + \tau^{(radio)}_{i,j} + \tau^{(pro)}_{i} \tag{8}$$

where $\tau^{(trans)}_{i,j}$, $\tau^{(radio)}_{i,j}$, and $\tau^{(pro)}_{i}$ are the transmission delay from node I to node J, the radio propagation delay from node I to node J, and the processing delay at the node I, respectively. The $\tau^{(radio)}_{i,j}$ is small enough to be ignored. The $\tau^{(trans)}_{i,j}$ is determined based on the bit rate of the channel and the data packet size. The $\tau^{(pro)}_{i}$ depends on the queuing type and switching technique used at each node. In some published research works, the $\tau^{(pro)}_{i}$ is usually determined based on the queue theory [13–16]. In this work, we estimate $\tau^{(pro)}_{i}$ for each packet as follows:

$$\tau^{(pro)}_{i} = t^{(out)}_{i} - t^{(in)}_{i} \tag{9}$$

where $t^{(in)}_{i}$ and $t^{(out)}_{i}$ are the time when the packet arrives at node I and the time when the packet leaves node I, respectively. The average EED of packets as they transmits through the hop $h_{i,j}$ in one period is given by

$$T_{i,j}(h) = \frac{\sum_{p=1}^{n} T^{(p)}_{i,j}(h)}{n} \tag{10}$$

where $n$ is the number of the packets transmitted from node I to node J in the period $t$. Same as [8], to avoid fluctuations, we use a coefficient $\beta \in [0, 1]$ that represents moving average for the $T_{i,j}(h)$. Thus, we have

$$T_{i,j}(h,t) = \beta T_{i,j}(h,t) + (1-\beta)T_{i,j}(h,t-1) \tag{11}$$

For the QGIR algorithm, the constraint of the EED is determined every time a node learns to update its routing table. Going back to the RL model of the QGIR algorithm in Fig. 1, when node I receives a hello packet from node J, node I updates the Q-value according to (13), where the learning rate factor $\alpha$ is determined by

$$\alpha = \begin{cases} m & if\ T_{i,j}(h,t) + T_{j,d}(r,t) \le T_{lim} \\ 0 & otherwise \end{cases} \quad (12)$$

where $m \in (0, 1]$, $T_{lim}$ is the limitation of the EED in the WMN to ensure the QoS. For the learning rate factor $\alpha$ is determined according to (12), In the case that the EED is greater than $T_{limit}$, the node I does not learn the new Q value. Therefore, the routes in the routing table of the node I always satisfy the EED constraint.

### 3.5. QGIR algorithm

To update the routing table at each node, the QGIR algorithm performs three routine tasks. One is that the nodes periodically send hello packets to provide their routing information to their neighbors (alg. 1), the other is that each node learns to update the routing table according to the Q-learning algorithm every time it receives a Hello packet (alg. 2), and the third is that each node learns by itself to update the routing table after each period of the PDP updates (alg. 3).

To advertise routing information to its neighbors, each node in the WMN periodically sends Hello packets for the neighbors to learn and update the routing information (alg. 1). To reduce overhead in the network, each node only sends hello packet whenever there is a change in routing information, the PDP of the hops to its neighbors (conditions at line 6 in alg. 1). In addition, to avoid looping, we use the split horizon technique. A node will not send a hello packet to a node that has previously learned the routing table.

Another proposal of us to reduce overhead packets in the network is the self-learning mechanism of the routing table at each node, the steps are as in alg. 3. Each node updates itself the routing table after each period of the PDP update of the hops in the network.

**Algorithm 1** Send Hello packets periodically at each node ($J$)

1: **while** (*true*) **do**
2: Wait($T_{helloLearn}$);
3: Read the best route from J to D in routing table of J;
4: **if** (Route $R_{j,k,d} = \{D, K, P_{j,k}(h), P_{j,k,d}(r)\}$ *is found*) **then**
5: Randomly select a node $I \in N_i, I \neq K$;
6: **if** $P_{i,j}(h) \neq P_{i,j}(h,l)$ **or** $P_{j,k,d}(r) \neq P_{j,k,d}(r,l)$ **then**
7: Create a Hello packet;
8: Store $P_{j,k,d}(r)$ and $T_{j,d}(r)$ into Hello packet;
9: Send Hello packet to node I;
10: **end if**
11: **end if**
12: **end while**

**Algorithm 2** Update Q-value at each node (I) when Hello packet is received.

1: Determine the IP address of the node that sends the Hello packet (J);
2: Read the information of $P_{j,k,d}(r)$ and $T_{j,d}(r)$ in the Hello packet;
3: Determine learning rate factor $\alpha$ according to (12);
4: Update Q-value and routing table according to (13);

**Algorithm 3** Each node (I) self-learns to update the routing table after each period $t$ of the PDP update.

1: **while** (*true*) **do**
2: wait($T_{selfLearn}$);
3: **for** (each node J is the neighbor of node I) **do**
4: Update $P_{i,j}(h,t)$ according to (3);
5: Update $T_{i,j}(h,t)$ according to (11);
6: Find $R_{i,j,d} = \{D, J, P_{i,j}(h), P_{i,j,d}(r)\}$ in routing table of I;
7: **if** ($R_{i,j,d}$ is found) **then**
8: $P_{i,j,d}(r) \leftarrow P_{i,j,d}(r) * P_{i,j}(h,t)/P_{i,j}(h)$;
9: $P_{i,j}(h) \leftarrow P_{i,j}(h,t)$;
10: Update $R_{i,j,d} = \{D, J, P_{i,j}(h), P_{i,j,d}(r)\}$;
11: **end if**
12: **end for**
13: **end while**

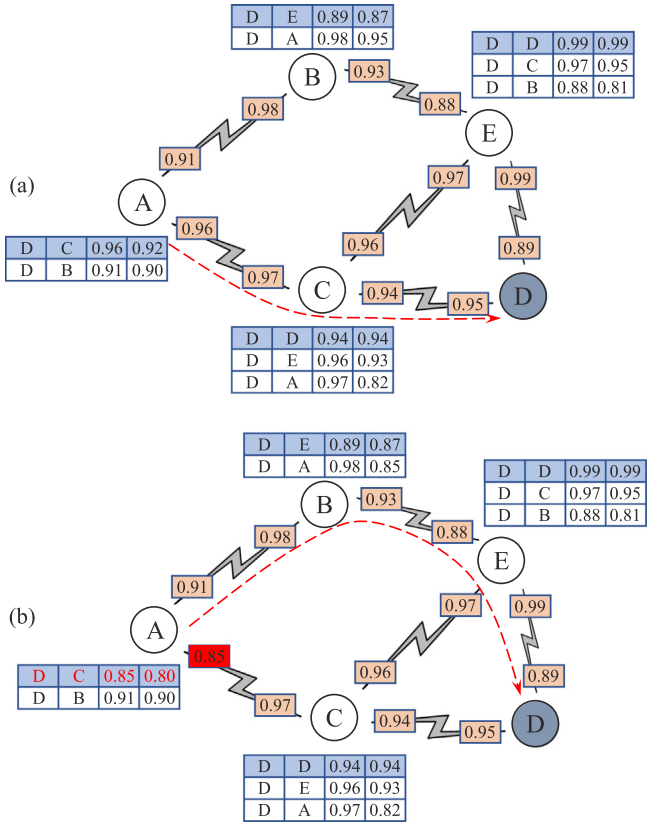



**Fig. 2.** An example of updating the routing table at each node using the QGIR algorithm.

To clearly see the principle of route discovery of the QALR algorithm, we consider an example as shown in Fig. 2, where the numerical value on each wireless link is the PDP over that hop at the time of consideration. Considering the routing table of each node to the D node for the case in Fig. 2(a). As defined in Table 1, the format of each route element from node J to node D is the $R_{j,d} = \{D, K, P_{j,k}(h), P_{j,d}(r)\}$. Consider the case that node A wants to transmit data to node D, the chosen

**Table 2**
Simulation parameters.

| Parameters | Setting |
|---|---|
| Simulation area | 1000 × 1000 (m) |
| Number of WR and gateway | 18 (nodes) |
| MAC protocol | 802.11ac |
| Modulation technique | 256-QAM |
| Data rate | 54 (Mbps) |
| Transmission range | 250 (m) |
| Traffic type | UDP |

route is $A \rightarrow C \rightarrow D$ because the PDP of this route is the best ($P_{A,D}(r) = 0.92$). Let us assume that after one period $t$, the PDP of hop $h_{A,C}$ reduces to 0.85 due to increased traffic load as shown in Fig. 2(b), Nodes send hello packets to their neighbors to update their routing tables according to Alg. 1. Consider the case where node C chooses node A to send hello packet, we show that the condition (6) in Alg. 1 is satisfied because $P_{A,C}(h) \neq P_{A,C}(h,l)$ ($P_{A,C}(h) = 0.85$, $P_{A,C}(h,l) = 0.96$), node C creates and sends hello packet to node A. When node A receives a Hello packet from node C, it will update the Q-value ($P_{A,C,D}(r)$ in our context) according to Alg. 2 to update into the routing table of node A. Assume the factors of the learning rate $\alpha$ and the discount $\gamma$ are set to 0.6 and 0.9, respectively, $P_{A,C,D}(r)$ is updated as follows

$$P_{A,C,D}(r) = (1-\alpha) \times P_{A,C,D}(r) + \alpha[P_{A,C}(h,t) \times \gamma \times \max_{\forall K \in L_C} P_{C,K,D}(r)] \quad (13)$$

$$= (1-0.6)0.92 + 0.6(0.85 \times 0.9 \times 0.94) = 0.80 \quad (14)$$

Thus, the first element in the routing table of the node A is updated to {D, C, 0.85, 0.80}. If at this time, node A transmits data to node B, the route redirects to $A \rightarrow B \rightarrow E \rightarrow D$, because the PDP of this route is greater than that of the route $A \rightarrow C \rightarrow D$.

## 4. Performance evaluation by simulation

In this section, we present the simulation results of the QGIR algorithm using OMNeT++ 5.6.2 [17] and INET Framework [18] to evaluate its performance. The QGIR algorithm is compared with the Destination Sequenced Distance Vector (DSDV) routing algorithm in terms of the PDP, EED, and throughput. The simulation assumptions are presented in Table 2. The learning rate factor ($m$ in Eq. (12)) and discount factor ($\gamma$ in Eq. (6)) have been set to 0.6, respectively.

In Fig. 3, we compare the performance of the DSDV and QGIR algorithms in terms of the PDP versus the network offered load. This is an important performance parameter of a network system. In our simulation scenario, the PDP is determined as the ratio of the total of the successfully received packets by the gateways to the total of sent packets by all source nodes in the network. The curves in Fig. 3 have shown that, when the traffic load is low, from 1000 to 3000 Kbps, the performances of both DSDV and QGIR algorithms in terms of the PDP are similar. This is because the QGIR chooses the route based on the reward function that is PDPs of the hops

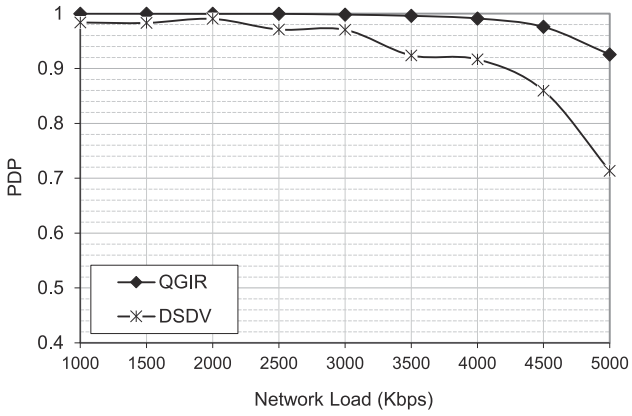



**Fig. 3.** Performance comparison of the QGIR and DSDV under packet delivery probability versus network load.

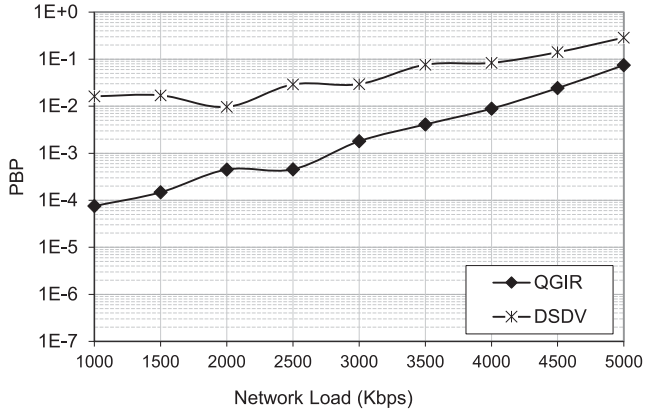



**Fig. 4.** Performance comparison of the QGIR and DSDV under packet blocking probability versus network load.

as described in Section 3.3. Meanwhile, with the low traffic load, the PDPs of the hops are very high, it can be close to 1. Therefore, choosing the route according to the PDPs metric in this case is equivalent to the hop count metric of the DSDV protocol. In the case of heavy traffic load, the PDP of the QGIR algorithm increases significantly compared with that of the DSDV algorithm. For example, considering the case that the traffic load is 3500 kbps, the PDPs of the DSDV and QGIR algorithm are 0.924 and 0.996, respectively. Thus, the PDP of the QGIR algorithm increases by 7.21% compared with that of the DSDV algorithm. For other cases, the higher the traffic load is, the more efficient algorithm QGIR is in terms of PDP.

Next, we analyze the PBP in the overall network, the obtained results are shown in Fig. 4, where we plot the PBP as a function of the network load. We can observe that, for the QGIR algorithm, PBP has reduced significantly compared with that of the DSDV algorithm, especially in the case of the heavy traffic load. Considering the case of the traffic load of 5000 Kbps, the PBP of DSDV and QGIR algorithms are $2.87 \times 10^{-1}$ and $7.44 \times 10^{-2}$, respectively. Thus PPB of the QGIR algorithm has reduced significantly 74.04% compared with that of the DSDV algorithm.

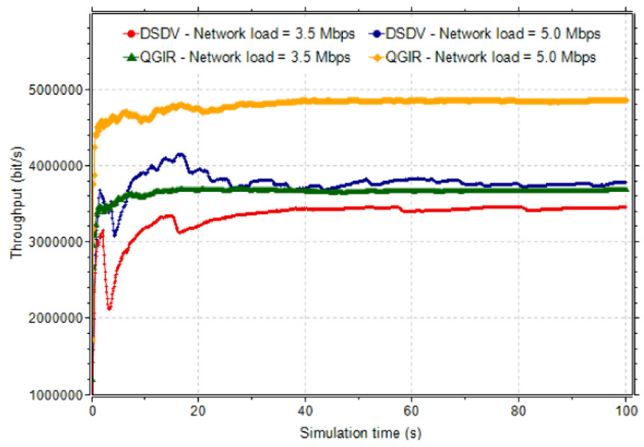



**Fig. 5.** Performance comparison of the QGIR and DSDV under network throughput.

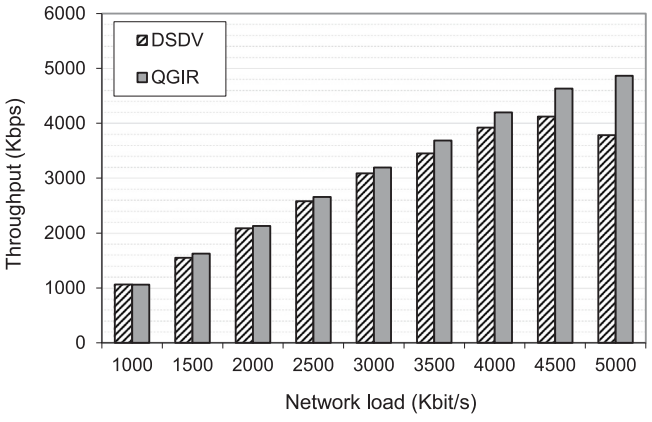



**Fig. 6.** Performance comparison of QGIR and DSDV under throughput versus network load.

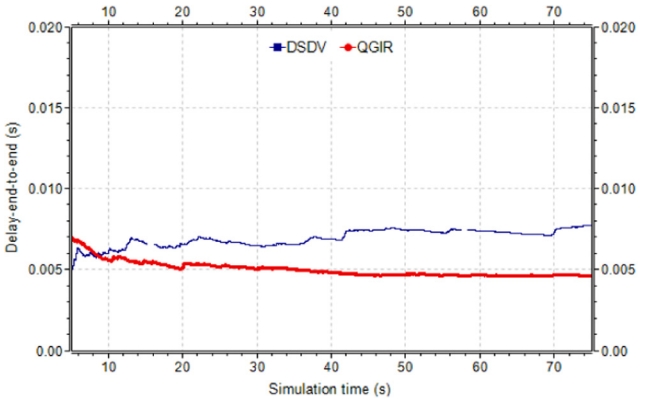



**Fig. 7.** Performance comparison of the QGIR and DSDV under delay-end-to-end.

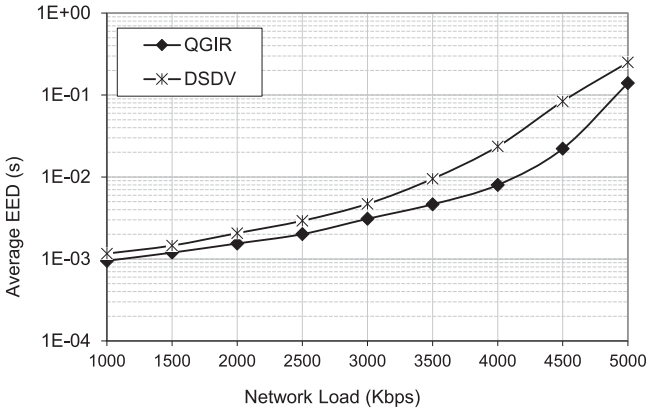



**Fig. 8.** Performance comparison of QGIR and DSDV under delay-end-to-end versus network load.

Another important performance parameter that also needs to be considered to ensure QoS is throughput. We also analyzed this parameter and obtained the results shown in Fig. 5, where we measure the network throughput versus simulation time for the cases that the traffic load is 3.5 and 5 Mbps. In our simulation, the throughput is defined as the successfully received traffic by the gateway in a time unit, measured in bits per second (bps) or its multiples. We can observe that the QGIR algorithm yields greater throughput than the DSDV algorithm in both cases of the network load of 3.5 and 5 Mbit/s. Specifically, considering the case of 5 Mbit/s, the average throughput of the DSDV and QGIR algorithms is 3.784 and 4.866 Mb/s, respectively. Thus, the average throughput of the QGIR algorithm has increased by 1.082 Mbit/s compared with that of the DSDV algorithm.

In case the traffic load changes, the average throughput of both DSDV and QGIR algorithms is shown in Fig. 6. We can observe that the higher the traffic load is, the higher the throughput is. However, the QGIR algorithm always yields greater throughput than the SPR algorithm.

Finally, we analyze the EED of both algorithms. The obtained results are shown in Fig. 7, where we measure the EED according to simulation time. In this case, the network load is 3500 Kbit/s. We can observe that, when the simulation time is less than 10 s, the average EED of both algorithms varies a lot due to the unstable network system. When simulation time is greater than 10 s, the average EED starts to stabilize. The QGIR algorithm A always yields a smaller average EED than the DSDV algorithm. This is caused by the QGIR algorithm reduces congestion at intermediate nodes, thus reducing the waiting time at the queues of these nodes, leading to a reduction in the EED. In a network model, the EED often depends mainly on the traffic load. This dependence we also analyzed in these simulation scenarios, the results are as shown in Fig. 8. We can observe that, for the QGIR algorithm, the average EED is always smaller than that of the DSDV algorithm, especially when the traffic load is higher, the difference is bigger.

## 5. Conclusions

The application of the RL for routing in wireless networks has attracted many research groups recently. The RL different methods can be effectively applied to improve routing protocols for the WMN. In this paper, we have proposed a QoS-guaranteed intelligent routing algorithm based on the RL for WMN with heavy traffic load, namely QGIR. The

simulation results have shown that the proposed algorithm has significantly improved the network performance in terms of the PDP, PBP, EED, and throughput compared with other well-known routing algorithms.

In the future, we will study the application of RL to control routing for other wireless network models, such as vehicular ad-hoc networks, flying ad hoc networks.

## CRediT authorship contribution statement

**Thuy-Van T. Duong:** Conception and design of study, Acquisition of data, Analysis and/or interpretation of data, Writing – original draft. **Le Huu Binh:** Conception and design of study, Acquisition of data, Writing – original draft, Writing – review & editing. **Vuong M. Ngo:** Analysis and/or interpretation of data, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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